

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

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13

Q1**B****B. 6**

Choice B is correct. It's given that Ty plans to walk at an average speed of 4 kilometers per hour. The number of kilometers Ty will walk is determined by the expression $4s$, where s is the number of hours Ty walks. The given goal of at least 24 kilometers means that the inequality $4s \geq 24$ represents the situation. Dividing both sides of this inequality by 4 gives $s \geq 6$, which corresponds to a minimum of 6 hours Ty must walk.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D****D. 0, 8**

Choice D is correct. The y -intercept of a line graphed in the xy -plane is the point where the line intersects the y -axis. The line graphed intersects the y -axis at the point 0, 8. Therefore, the y -intercept of the line graphed is 0, 8.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q3**8**

The correct answer is 8. Adding x to both sides of the given equation yields $14x = 112$. Dividing both sides of this equation by 14 yields $x = 8$.

Q4**C****C.** 0, 2

Choice C is correct. The y -intercept of a graph is the point where the graph intersects the y -axis. The graph of $y = f(x)$ shown intersects the y -axis at the point $(0, 2)$. Therefore, the y -intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the x -intercept, not the y -intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**A****A.** 15, 3

Choice A is correct. The second equation in the given system defines the value of x as $5y$. Substituting $5y$ for x into the first equation yields $5y + y = 18$ or $6y = 18$. Dividing each side of this equation by 6 yields $y = 3$. Substituting 3 for y in the second equation yields $5(3) = x$ or $x = 15$. Therefore, the solution x, y to the given system of equations is 15, 3.

Choice B is incorrect. Substituting 16 for x and 2 for y in the second equation yields $5(2) = 16$, which is not true. Therefore, 16, 2 is not a solution to the given system of equations.

Choice C is incorrect. Substituting 17 for x and 1 for y in the second equation yields $5(1) = 17$, which is not true. Therefore, 17, 1 is not a solution to the given system of equations.

Choice D is incorrect. Substituting 18 for x and 0 for y in the second equation yields $5(0) = 18$, which is not true. Therefore, 18, 0 is not a solution to the given system of equations.

Q6**A****A.** 1,974

Choice A is correct. It's given that a student's essay needs to have no more than 5,000 words. It's also given that the student's essay already has 3,026 words. Let x represent the number of additional words the student includes in the essay. After including x additional words, the essay will have a total of $3,026 + x$ words. Since the essay needs to have no more than 5,000 words, it follows that $3,026 + x \leq 5,000$. Subtracting 3,026 from both sides of this inequality yields $x \leq 1,974$. Thus, the maximum number of additional words the student can include in the essay is 1,974.

Choice B is incorrect. This is the number of words already in the student's essay, rather than the maximum number of additional words the student can include.

Choice C is incorrect. This is the total number of words the essay can contain, rather than the maximum number of additional words the student can include.

Choice D is incorrect. This is the number of words the essay would contain if the student added 5,000 more words, rather than the maximum number of additional words the student can include.

Q7**A**

A. $y = -\frac{1}{4}x - 1$

Choice A is correct. The slope of the line can be found by choosing any two points on the line, such as $(4, -2)$ and $(0, -1)$. Subtracting the y-values results in $-2 - (-1) = -1$, the change in y. Subtracting the x-values results in $4 - 0 = 4$, the change in x. Dividing the change in y by the change in x yields $-1 \div 4 = -\frac{1}{4}$, the slope. The line intersects the y-axis at $(0, -1)$, so -1 is the y-coordinate of the y-intercept. This information can be expressed in slope-intercept form as the equation $y = -\frac{1}{4}x - 1$.

Choice B is incorrect and may result from incorrectly calculating the slope and then misidentifying the slope as the y-intercept. Choice C is incorrect and may result from misidentifying the slope as the y-intercept. Choice D is incorrect and may result from incorrectly calculating the slope.

Q8**480**

The correct answer is 480. Multiplying both sides of the given equation by 8 yields $82 + x = 860$, or $16 + 8x = 480$. Therefore, if $2 + x = 60$, the value of $16 + 8x$ is 480.

Q9**C**

C. $c = 11d + 10$

Choice C is correct. It's given that the cost of renting a tent is \$ 11 per day for d days. Multiplying the rental cost by the number of days yields \$ $11d$, which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of \$ 10 to the rental cost of \$ $11d$ gives the total cost c , in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was \$ $d + 11$ for 10 days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was \$ 10 and the onetime fee was \$ 11.

Q10**A**

A. 8,9

Choice A is correct. The second equation in the given system is $x = 8$. Substituting 8 for x in the first equation in the given system yields $4(8) - 3y = 5$, or $32 - 3y = 5$. Subtracting 32 from both sides of this equation yields $-3y = -27$. Dividing both sides of this equation by -3 yields $y = 9$. Therefore, the solution (x, y) to the given system of equations is $(8, 9)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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